

Semi-Humanoid Robotics: A Comprehensive Literature Review of Platforms, Control, Embodied AI and Systems Integration

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Abstract

Semi-humanoid robotics joins a wheeled or tracked mobile base, a torso or lift, articulated arms, compliant end effectors and a sensorized interaction head. This report comprehensively consolidates two supplied literature reviews: an arXiv-oriented review of emerging open platforms, teleoperation and embodied-AI systems, and an OpenAlex-oriented review of peer-reviewed work on tactile sensing, task orchestration, human–robot interaction (HRI), digital twins and mobile manipulation. The reviewed evidence supports a systems research agenda in which the main challenge is not any isolated subsystem but the validated integration of modular mechanics, robust power and data interfaces, tactile contact, real-time safety, recovery logic, simulation and human-centred communication. The report separates source-reported findings from proposed design implications, retains all cited works and source-specific claims, and includes verbatim archival appendices of both supplied Markdown reports. Thus, no wording from the input dossiers is omitted from the report record.

Keywords: semi-humanoid robotics; mobile manipulation; bimanual teleoperation; tactile sensing; behavior trees; vision-language-action; digital twins; ROS 2; systems integration.

1 Introduction

Research in robotics is moving from isolated capabilities—fixed-base manipulation, navigation, or bipedal locomotion—toward integrated systems that must move, perceive, manipulate and safely interact with people. The supplied arXiv dossier characterises a semi-humanoid as a wheeled/track base, vertical torso or lift, dual 6–7 DoF arms, tactile or under-actuated grippers and an expressive sensor head. The supplied OpenAlex dossier complements this systems view with peer-reviewed foundations in compliant tactile sensing, behavior-tree reactivity, trajectory primitives, proactive HRI and human-centric digital twins.[2, 1, 14, 18]

This report has two purposes. First, it provides a research-paper-form synthesis that preserves each substantive strand in the two dossiers. Second, it establishes a complete audit trail: Appendices ?? and ?? reproduce the supplied source documents verbatim. The appendices are archival evidence; the main text uses typeset equations, tables and editable TikZ diagrams rather than reproducing the sources’ ASCII diagrams.

2 Methodology and evidence protocol

2.1 Corpus and scope

The corpus consists exclusively of the two user-provided Markdown reports. Both describe developments from 2022–2026. The arXiv review foregrounds emerging preprints, open platforms and implementation patterns. The OpenAlex review foregrounds peer-reviewed journals, conferences, DOIs and bibliometric claims. No new external search or independent source validation was undertaken for this synthesis.

2.2 Synthesis procedure

Each source was mapped into six common dimensions: platform architecture; manipulation and tactile hardware; teleoperation and whole-body control; task orchestration; multimodal AI and HRI; and simulation/digital twins. Repeated claims were reconciled, source-specific details were retained, and design recommendations were translated into falsifiable experimental artefacts. Citations refer to the works as listed in the supplied dossiers.

2.3 Evidence limits

The report uses “reported” for values, costs, citation counts, success rates and timing figures drawn from the source dossiers. Preprints offer timely system insight but should not be treated as peer-reviewed validation. Conversely, peer-reviewed component or survey literature does not independently demonstrate the complete

proposed platform. Claims of “gaps” are therefore framed as gaps in the provided corpus or as integration opportunities, not absolute proof that no prior work exists.

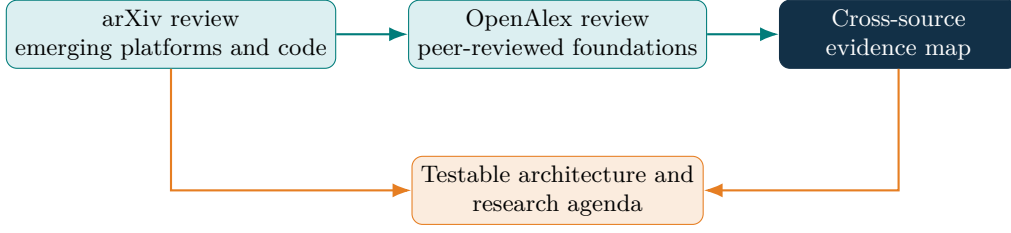


Figure 1: Evidence protocol: emerging and peer-reviewed evidence are integrated, but their evidentiary roles remain distinct.

3 Taxonomy of semi-humanoid systems

3.1 Form factor and whole-body state

For structured indoor environments, wheeled or omnidirectional mobility can reduce the cost and stability burden of bipedal locomotion while retaining mobile access to human-designed spaces. Mecanum or swerve bases provide planar x, y, θ motion and zero-radius turning; a telescopic lift substitutes vertical workspace expansion for leg kinematics. The whole-body state can be expressed as

$$\mathbf{q} = \begin{bmatrix} \mathbf{q}_b^T & q_\ell & \mathbf{q}_L^T & \mathbf{q}_R^T \end{bmatrix}^T, \quad \dot{\mathbf{x}}_e = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}}, \quad (1)$$

where $\mathbf{q}_b$ describes base motion, q_ℓ lift displacement, and $\mathbf{q}_L, \mathbf{q}_R$ arm joints. Whole-body inverse kinematics and MPC may coordinate these variables to preserve stability and reach.[5]

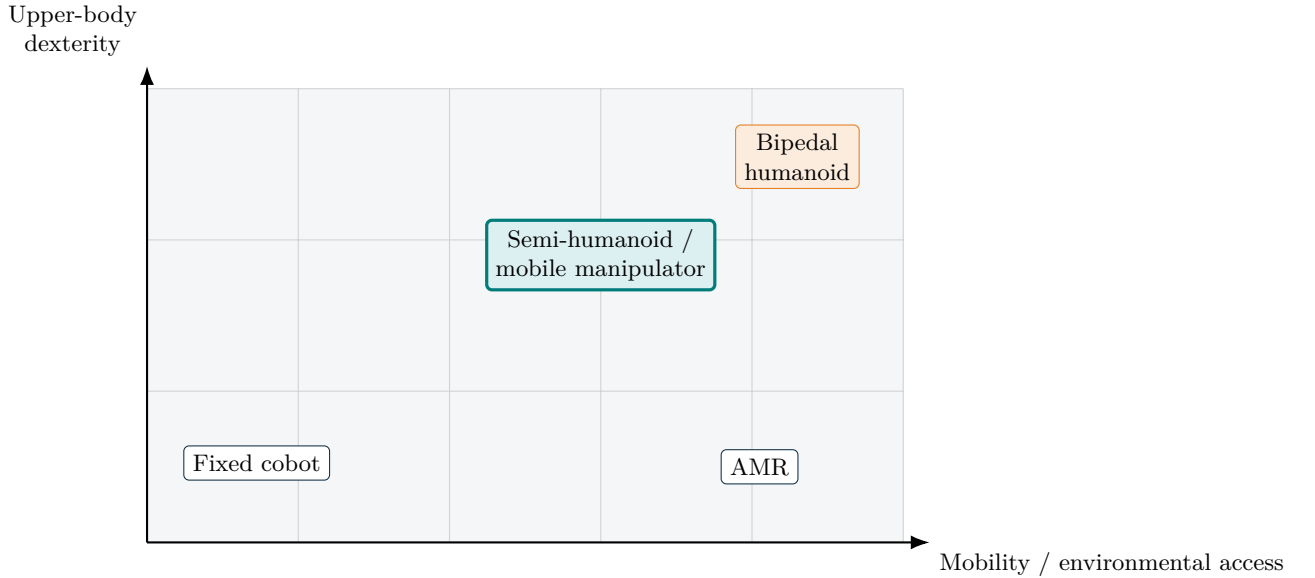


Figure 2: Design-space position of a semi-humanoid platform.

3.2 Platform benchmark and architectures

Table 1 retains the platform benchmark reported in the arXiv review. The benchmark mixes platform generations, openness models and cost-estimation bases; it is consequently a design reference rather than a controlled comparison.

Table 1: Platform taxonomy and reported comparison from the supplied arXiv review.

Platform	Base and torso/reach	Arms, hands and HRI	Reported BOM	Hardware/software model
YOR (NYU/Meta, 2026)	Omnidirectional; telescopic vertical lift	Dual 6-DoF arms, parallel grippers; dual RGB-D/head mount	<\$10k	Open source, COTS
Mobile ALOHA (Stanford, 2024)	Tracer wheeled chassis; fixed frame	Dual 6-DoF ViperX + leader arms; wrist/top RGB	\$~32k	Open hardware / software
Stretch 3 (Hello Robot)	Differential drive; telescopic column	1-DoF arm + 2-DoF wrist gripper; ReSense RGB-D	\$~25k	Commercial, open API
TiAGO (PAL Robotics)	Differential/omni; motorised spine lift	Single/dual 7-DoF arms; pan-tilt head + microphone array	>\$90k	Commercial / proprietary
PR2 (Willow Garage)	Omnidirectional; telescopic spine	Dual 7-DoF arms; pan-tilt, stereo, LiDAR	>\$280k	Open source legacy baseline
Proposed platform	Modular omni/Mecanum; telescopic spine/active tilt	Dual modular 6–7 DoF; 3-DoF head, OAK-D, mic array	\$12–\$18k	Fully open integration standard

Evidence basis: Supplied arXiv review; cited platform works include [1, 2, 3].

4 Comprehensive technical synthesis

4.1 Pillar 1: modular mechatronics, mobility and bimanual platforms

The arXiv dossier reports that bipedal locomotion can add unnecessary dynamic instability and cost in structured indoor settings. It highlights 3-DoF planar omnidirectional mobility, telescopic lift columns reaching floor to counter height, and low distal inertia through shoulder-mounted motors, Dyneema tendons and Bowden routing. It also identifies passive/adaptive end-effectors such as LEAP and BitoHand, with tendon networks and fingertip FSR arrays.[1, 3]

The OpenAlex dossier complements these platform patterns with RSS 2023 low-cost bimanual hardware and imitation learning, reported as achieving over 90% task success across fine-grained tasks such as bag zipping and bottle opening. It also includes a 2022 Precision Agriculture survey, which motivates wide-field RGB-D navigation plus wrist-mounted visual servoing in cluttered greenhouse picking.[11, 17]

4.2 Pillar 2: tactile hands, soft-rigid mechanisms and contact sensing

The scholarly corpus contains a soft, thumb-sized vision-based tactile sensor for continuous 3D force and sub-millimetre deformation estimation, a soft-rigid compliant-actuation review, an iontronic pressure-sensor reference and reports on tactile/soft hand concepts.[12, 13, 20] Together, these works motivate a multi-scale sensing topology: a wrist 6-axis force/torque sensor for global load, fingertip FSR or pressure sensing for local contact, and visual/tactile imagery for deformation and slip cues. A generic contact-state estimator is

$$\hat{\mathbf{c}} = f_{\theta}(\mathbf{F}_{\text{wrist}}, \mathbf{p}_{\text{finger}}, \mathbf{I}_{\text{tactile}}, \mathbf{I}_{\text{vision}}). \quad (2)$$

The input documents identify an unresolved integration issue: published systems rarely standardise a modular wrist-to-fingertip topology combining rigid F/T sensing with compliant silicone skins.

4.3 Pillar 3: teleoperation, whole-body control, behavior trees and DMPs

Mobile ALOHA and OmniH2O are identified as leader-follower/master-slave infrastructures for whole-body teleoperation and data capture.[2, 4] The arXiv dossier also reports ROS 2 Humble/Jazzy, Zenoh and

micro-ROS as an evolution toward low-latency middleware and embedded execution. The OpenAlex dossier adds the Behavior Tree survey and the DMP tutorial survey: BTs provide modularity, reactivity and fault recovery compared with conventional FSMs, while DMPs provide a mathematically grounded way to represent, adapt and scale demonstrated trajectories.[14, 15]

The combined implication is a two-part control architecture: a task layer that selects, sequences and recovers skills, and a fast control layer that regulates motion and contact. Bimanual operations need explicit recovery nodes for partial slip, blocked paths, elevator/door interaction, object handover and safe retreat—an open library and evaluated benchmark are more valuable than a one-off task graph.

4.4 Pillar 4: multimodal AI and expressive HRI

The arXiv report lists OpenVLA and SLIM as open VLA/action-grounded model directions that translate language-conditioned visual information into end-effector action proposals.[6, 7] It also records work on expressive robotic heads, gaze/attention, episodic memory and TDOA microphone-array localisation.[8, 9] The OpenAlex review provides a peer-reviewed HRI foundation: proactive, predictable collaboration depends on intention signalling, including head orientation, gaze and displays before physical movement.[16]

The proposed 3-DoF pan-tilt-roll head with OAK-D stereo vision, four-microphone TDOA array and OLED face display is therefore an experimentally testable HRI hypothesis. It should be assessed using predictability, workload, trust, attention alignment and perceived safety metrics rather than asserted to improve interaction by default.

4.5 Pillar 5: simulation, digital twins and sim-to-real

The source documents name MuJoCo, Isaac Sim, OpenUSD, semantic 3D Gaussian Splatting and reinforcement learning as simulation and pre-training infrastructure.[10, 18, 19] The OpenAlex review foregrounds human-centric industrial digital twins that join physical telemetry, scene graphs and synthetic data for monitoring and collision prediction. The stated limitation is tactile/deformable contact: rigid-body simulation may represent chassis and arm kinematics well but does not necessarily reproduce silicone skin deformation, FSR response or contact friction. A credible pipeline must quantify the discrepancy:

$$\epsilon_{\text{sim2real}} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{y}_i^{\text{sim}} - \mathbf{y}_i^{\text{real}}\|, \quad (3)$$

for declared output variables $\mathbf{y}$ such as force, pose, slip classification or task success.

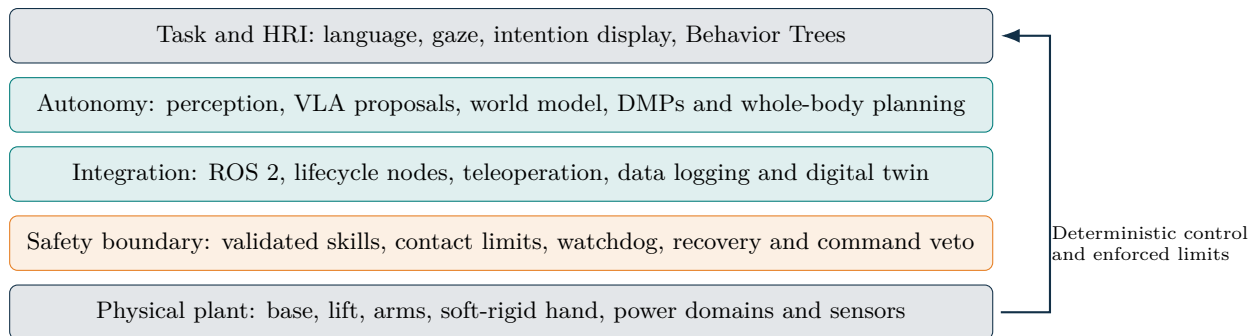


Figure 3: Integrated architecture implied by both dossiers. The orange boundary separates non-deterministic high-level proposals from real-time safety enforcement.

5 Gap analysis and research contributions

The two dossiers collectively identify ten gap statements. Five recur across both sources and five are source-specific elaborations. Table 2 retains all of them while converting them into validation-oriented contributions.

Table 2: Complete gap-to-contribution map from the supplied reviews.

Gap	Source-reported problem	Research-grade contribution and evidence
Modular mechanical/electrical interfaces	Fixed, monolithic builds require manual rewiring and hard-coded reconfiguration when swapping arms, grippers or torso/base configurations.	ISO 9409-1-compatible quick-release pattern, blind-mate power/CAN, ROS 2 lifecycle discovery; report swap time, fault rate and configuration reproducibility.
VLM/VLA-to-motor latency and safety chasm	Source reports VLA inference at 5–10Hz versus contact-safe control at 1kHz; raw trajectory execution lacks a standard reflex veto.	Jetson cognitive layer at 10–30Hz; STM32/ESP32 micro-ROS layer at 1kHz; command-veto, contact and communication-loss fault-injection tests.
Galvanic clean/dirty power isolation	Back-EMF and high-current actuation can reset Jetson/camera logic; papers often omit PDB/schematic standards.	Isolated 24–48V motor and regulated 12/19V compute domains; transient, EMC, brownout and reset-resilience measurements.
Static/non-expressive low-cost heads	Mobile ALOHA/YOR-style static camera posts do not provide social cues, speaker tracking or integrated intention display.	Open 3-DoF head with OAK-D, four-mic TDOA and OLED face; compare HRI measures against a static-head baseline.
Cost, proprietary lock-in and fragility	Commercial platforms are reported at \$90k–\$280k while DIY platforms lack durability, documentation and replaceable-module standards.	Publish interface documents, BOM, calibration and service procedures; measure total build, repair and replication cost.
Rigid load-cell/soft-skin disconnect	Rigid 6-axis wrist sensors and soft skins are seldom integrated in one modular sensing topology.	Calibrated wrist-to-fingertip sensing stack; report contact-transition and load-sharing performance.
No standard bimanual BT recovery nodes	Existing BT implementations focus on single-arm or navigation cases.	Open BehaviorTree.CPP nodes for slip, obstruction, handover, door/elevator and retreat; report recovery success and time-to-recovery.
Deformable tactile sim-to-real gap	Isaac Sim-style rigid-body workflows do not fully model silicone deformation and fingertip FSR contact physics.	Instrumented contact benchmark, domain randomisation, hardware/simulation error curves and transfer protocol.
Agricultural/mobile-manipulator perception challenge	Cluttered unstructured picking needs broad navigation sensing plus wrist visual servoing.	Compare head, base and wrist sensor placement on clutter/occlusion tasks.
Data and demonstration cost	Real-world data collection is expensive, motivating teleoperation, synthetic data and policy pre-training.	Teleoperation dataset schema, quality controls, task coverage metrics and simulator-assisted augmentation study.

6 System design recommendations retained from the source dossiers

6.1 Hardware and interface recommendations

The source recommendations call for a plug-and-play modular flange architecture: ISO 9409-1 mechanical patterns, combined power/CAN connectors and automatic ROS 2 lifecycle discovery. They propose modular SO-ARM100/Koch arms, LEAP/tactile hands, VESC motor controllers and a target BOM below \$15k (with a broader stated estimate of \$12k–\$18k). Proximal tendon actuation is recommended to lower distal inertia;

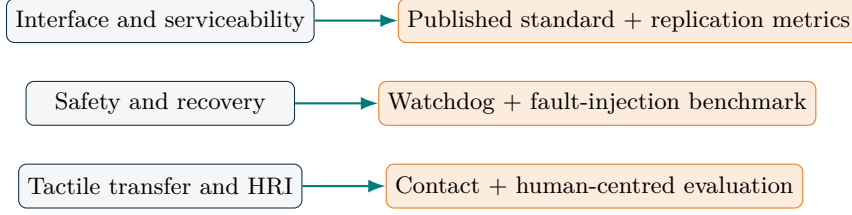


Figure 4: A research contribution should convert the identified integration gaps into measurable artefacts and reproducible evidence.

soft-rigid five-finger hands combine wrist F/T sensing with fingertip FSRs.

6.2 Compute, safety and power recommendations

Both reports recommend a “brain and spinal cord” architecture: Jetson Orin with VLM/LLM and ROS 2 at approximately 10–30Hz, plus STM32/ESP32 micro-ROS at 1kHz for deterministic execution and impedance/reflex enforcement. At the actuator boundary, a generic impedance controller is

$$\mathbf{F}_{\text{cmd}} = \mathbf{K}(\mathbf{x}_d - \mathbf{x}) + \mathbf{D}(\dot{\mathbf{x}}_d - \dot{\mathbf{x}}), \quad (4)$$

with independent limits for force, velocity, temperature and loss of communication. The supplied sources further recommend optocouplers/galvanic isolation and a dual-domain PDB separating dirty actuator power from clean compute/sensor power.

6.3 Software, HRI and digital-twin recommendations

The retained software proposal is ROS 2 with MoveIt 2/`ros2_control`, BehaviorTree.CPP recovery nodes, teleoperation data capture, and an Isaac Sim OpenUSD SIL/vHIL pipeline with domain randomisation. The HRI proposal is a 3-DoF gimbal head integrating OAK-D stereo vision, a four-mic TDOA array and an OLED expression screen. The sources link these choices to project references including strawberry harvesting, FLEXIBOT, multi-floor navigation, WMS forklift operation, soft-rigid hand and empowered grasp projects; their complete mappings appear verbatim in Appendix ??.

7 Research methodology for validation

7.1 Experimental questions

The combined literature suggests five answerable research questions: (RQ1) Does the interface standard reduce module-change time and integration errors? (RQ2) Does the enforced safety layer reject unsafe high-level commands within the specified control period? (RQ3) Does sensor fusion improve contact transition and grasp retention across compliant objects? (RQ4) Do bimanual recovery nodes improve completion and recovery time relative to a nominal-only tree? (RQ5) Does an active perception head improve specified HRI outcomes versus a static baseline?

7.2 Metrics and reporting

Report task success, productive-cycle time, intervention rate, recovery success, mean time to recovery, module swap time, availability, energy use, sensor/compute resets, contact-force tracking error and $\epsilon_{\text{sim2real}}$. For HRI, predefine participant population, task, counterbalancing, questionnaire/scales and safety protocol. For all benchmarks, publish configurations, object sets, calibration, random seeds, failure cases and source code where possible.

7.3 Threats to validity

Reported prices and platform specifications are configuration-dependent. Teleoperation results may not generalise from demonstrations to autonomous execution. Simulation may conceal contact, latency and electrical effects. Small HRI studies may overstate generality. These threats reinforce the need for modular baselines, transparent failure reporting and on-hardware tests.

8 Conclusion

The supplied arXiv and OpenAlex reviews jointly describe a coherent field: mobile manipulation hardware is becoming more accessible; teleoperation and demonstrations are reducing data barriers; tactile sensing, BTs, DMPs and digital twins provide mature component foundations; and multimodal AI is expanding the space of high-level interaction. The strongest semi-humanoid research contribution is not a claim of general intelligence. It is a complete, measurable integration: modular and serviceable hardware, isolated power, explicit safety boundaries, calibrated contact sensing, recoverable task execution, validated sim-to-real transfer and human-centred communication.

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